Transceiver Architecture & Channel Model

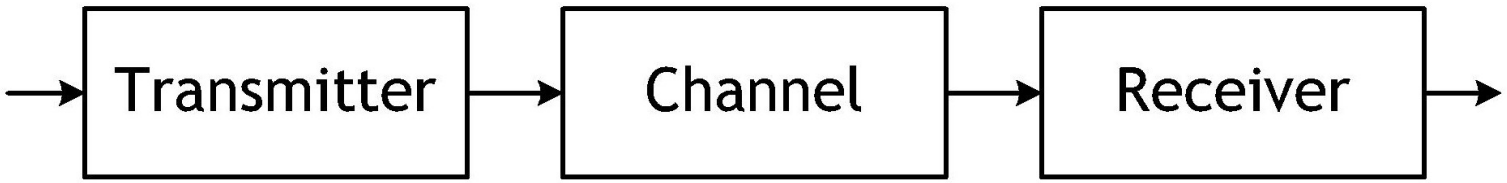
Instructor : Pei-Yun Tsai

Digital Communication IC Design

"Baseband Receiver Design for Wireless MIMO-OFDM Communications" 2nd Edition, Wiley 2012.
Chapter 5, Signal Propagation and Channel Model

Communication Systems

- Transmitter: transmit information by a continuous-time signal.
- Channel: physical medium for communication.
- Receiver: recover the information via signal processing techniques.

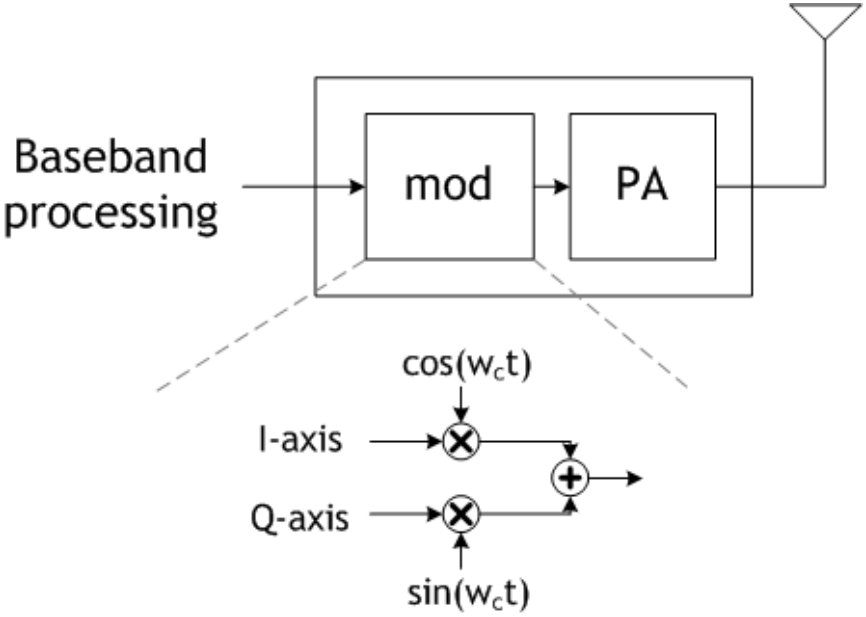


Circuits Blocks in Transceiver

- Transmitter
  - Power amplifier, pre-amplifier, mixer
  - Filter
  - Baseband processor
- Receiver
  - LNA, mixer, variable gain amplifier (VGA)
  - Filter
  - Baseband processor
- Frequency synthesizer
  - Phase-locked loop
  - Phase/Frequency detector (PFD), VCO, divider
- ADC/DAC

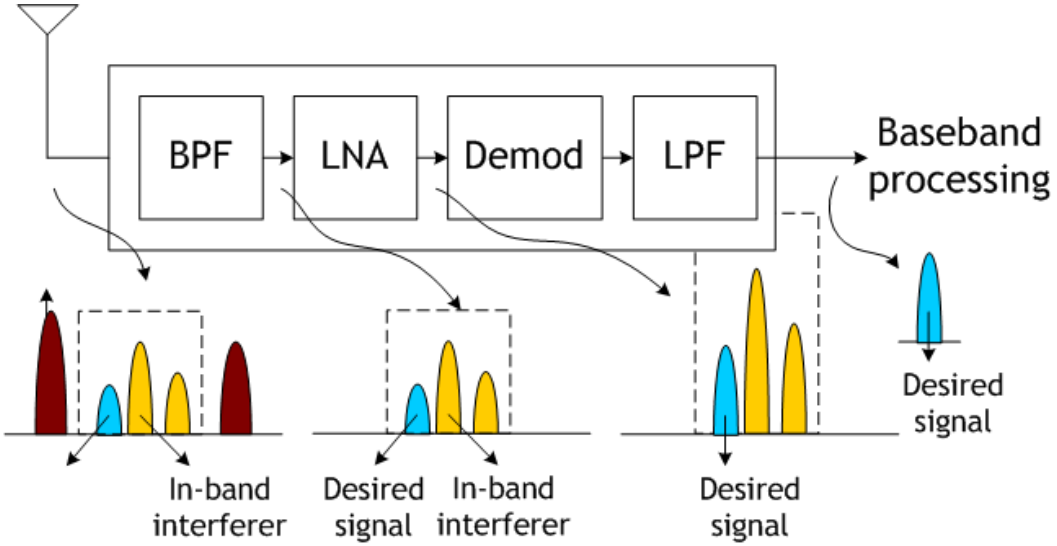
Considerations (1/2)

- For transmitter
  - Modulation Scheme
    - Constant Envelope v.s. variable envelope
    - Spec. of PA design
  - Tradeoffs between spectral efficiency and power efficiency
  - IQ mismatch causes crosstalk between I and Q axes.



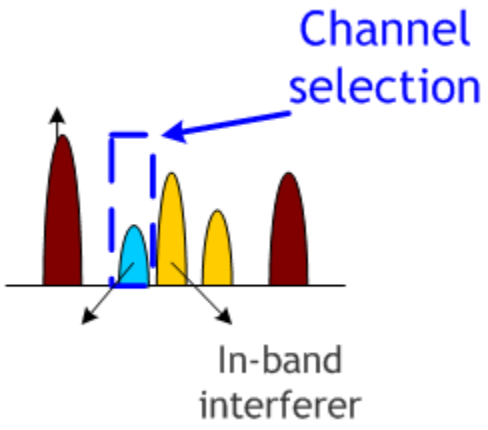
Considerations (2/2)

- For receiver
  - Small desired signal (~-100dBm)
    - Solution: Amplify
  - Large in-band/out-of-band interferer (~-20dBm)
    - Solution: Filter
  - Noise and distortion added in building blocks
    - Solution: Noise limitation

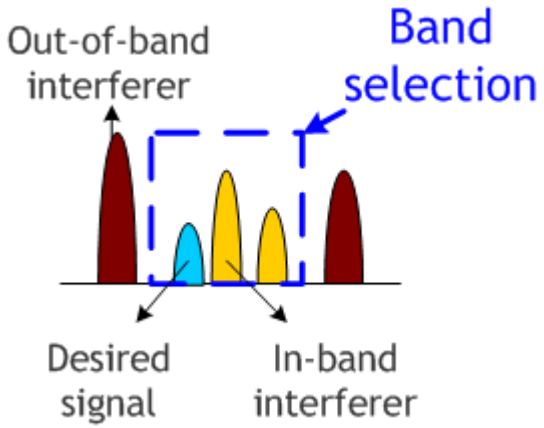


Filtering

- Front-end channel selection may be difficult
  - High Q band-pass filter
  - $Q=W_c/BW=950\ MHz/\ 200KHz=4750$

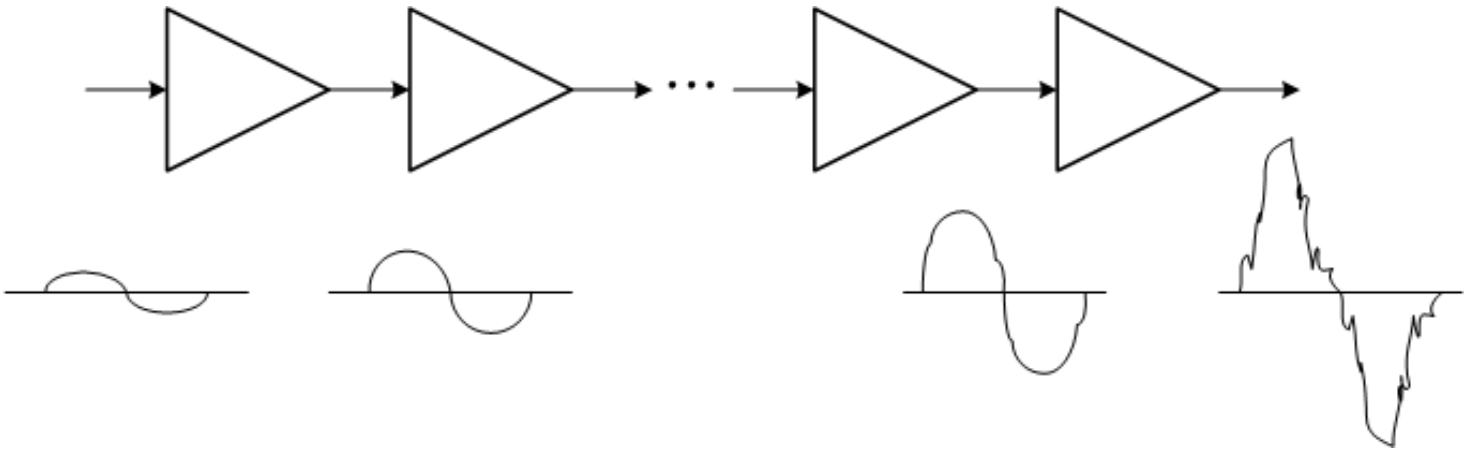


- Band selection first to reject out-of-band interferer and then channel selection after demodulation
  - Practical Q band-pass filter
  - $Q=W_c/BW=950\ MHz/\ 25MHz=38$



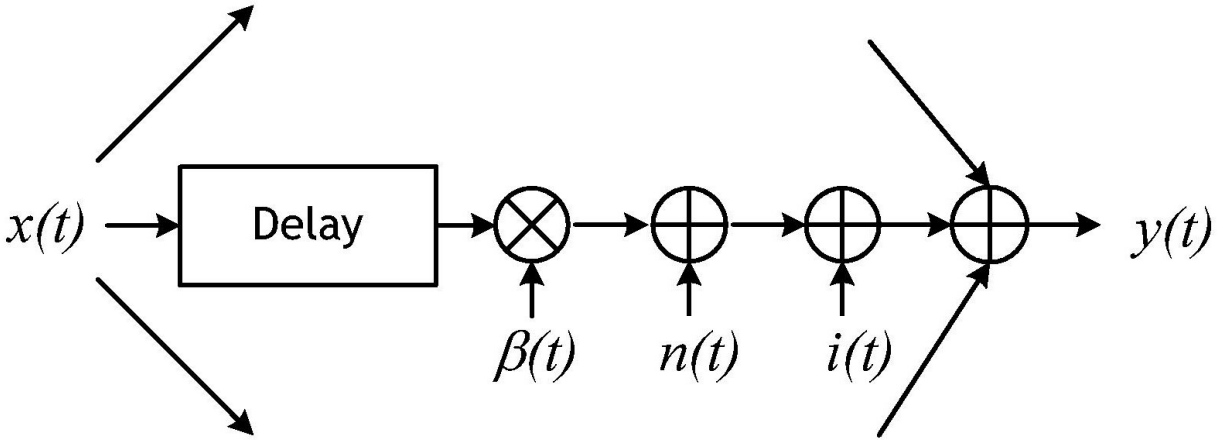
Amplifying

- Small signal needs to be boosted up by cascading finite-gain amplifiers.
- Each amplifier must be design appropriately to minimize additional noise and distortion.



Physical Channel

- Channel model provides statistical properties.
- Real channel is very complicated.



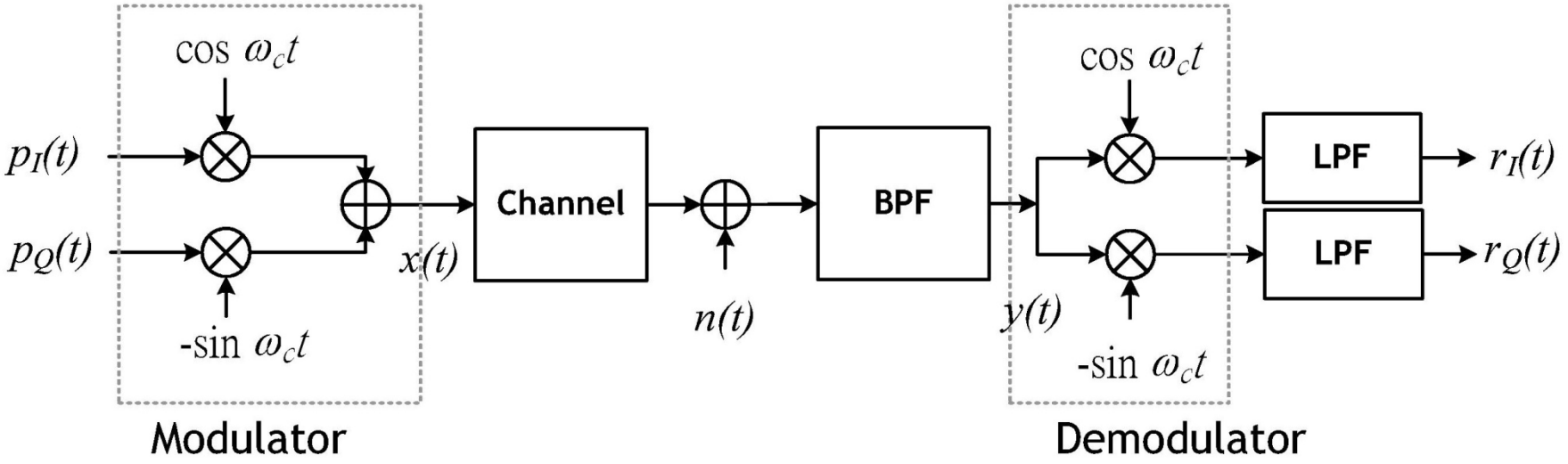
Wireless Signal Propagation

Baseband Equivalent Channel (1/5)

- Baseband signal processing aims to deal with baseband equivalent channels.

$$x(t) = p_I(t)\cos(\omega_c t) - p_Q(t)\sin(\omega_c t) = \mathrm{Re}\{\left(p_I(t) + jp_Q(t)\right)e^{j2\pi f_c t}\}$$

$$y(t) = (x(t) \otimes h_c(t) + n(t)) \otimes h_{BPF}(t)$$

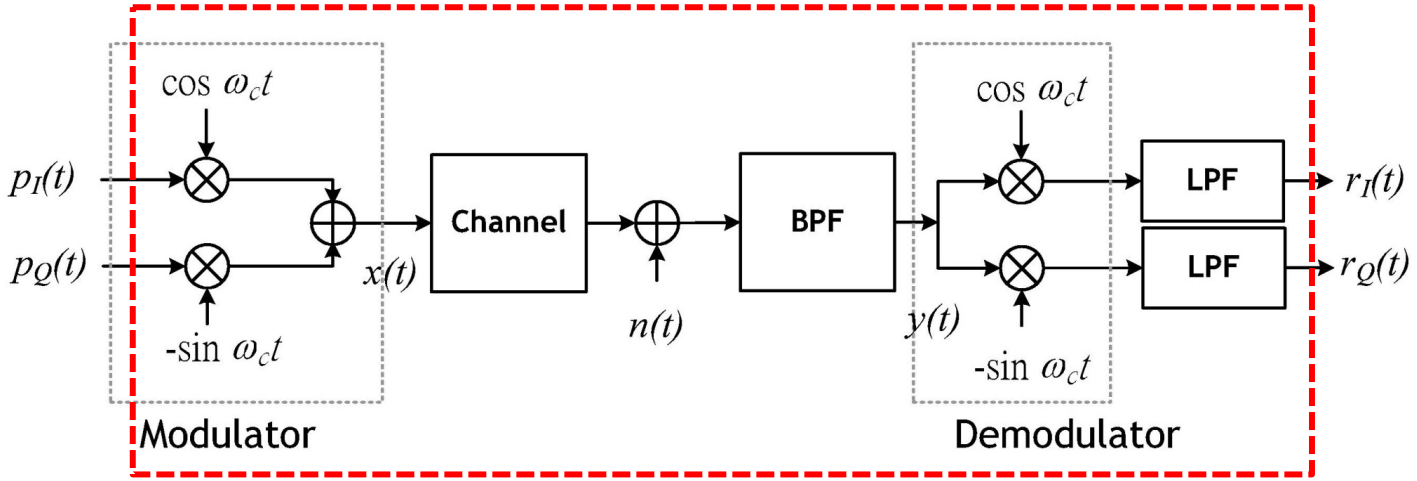


$$r_I(t) = (y(t)\cos(\omega_c t)) \otimes h_{LPF}(t)$$

$$r_Q(t) = (-y(t)\sin(\omega_c t)) \otimes h_{LPF}(t)$$

$$r(t) = (r_I(t) + jr_Q(t)) = (y(t)e^{-j2\pi f_c t}) \otimes h_{LPF}(t)$$

Baseband Equivalent Channel (2/5)



$$r(t) = (r_I(t) + jr_Q(t)) = \quad ? \quad \left(p_I(t) + jp_Q(t)\right)$$

Ignore noise

$$y(t) = \left(x(t) \otimes h_c(t)\right) \otimes h_{BPF}(t) = \mathrm{Re}\left\{\left(p_I(t) + jp_Q(t)\right) e^{j2\pi f_c t}\right\} \otimes h_c(t) \otimes h_{BPF}(t)$$

$$r(t) = \left(y(t) e^{-j2\pi f_c t}\right) \otimes h_{LPF}(t)$$

$$= h_{LPF}(t) \otimes \left[e^{-j2\pi f_c t} \left(h_{BPF}(t) \otimes h_c(t) \otimes \mathrm{Re}\left\{\left(p_I(t) + jp_Q(t)\right) e^{j2\pi f_c t}\right\}\right) \right]$$

Baseband Equivalent Channel (3/5)

- Given $(p_I(t) + jp_Q(t)) \leftrightarrow P(f)$
  - $\mathrm{Re}\left\{\left(p_I(t) + jp_Q(t)\right)e^{j2\pi f_c t}\right\}$

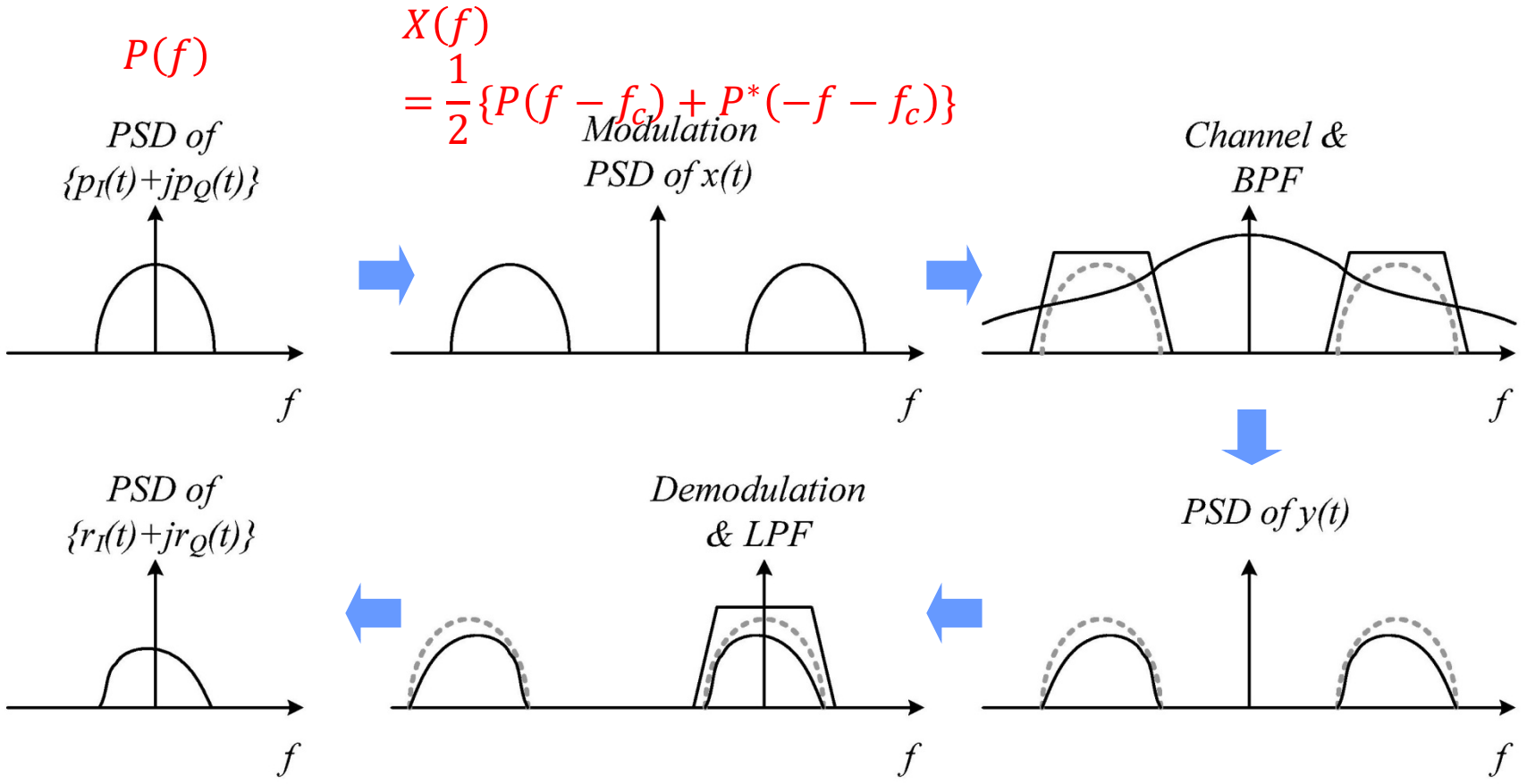
    $= \frac{1}{2}\left\{\left(p_I(t) + jp_Q(t)\right)e^{j2\pi f_c t} + \left(p_I(t) + jp_Q(t)\right)^* e^{-j2\pi f_c t}\right\}$

    $\leftrightarrow \frac{1}{2}\{P(f - f_c) + P^*(-f - f_c)\}$

- Given $(y(t)) \leftrightarrow Y(f)$
  - $(y(t)e^{-j2\pi f_c t}) \leftrightarrow Y(f + f_c)$

Baseband Equivalent Channel (4/5)

- Frequency domain response



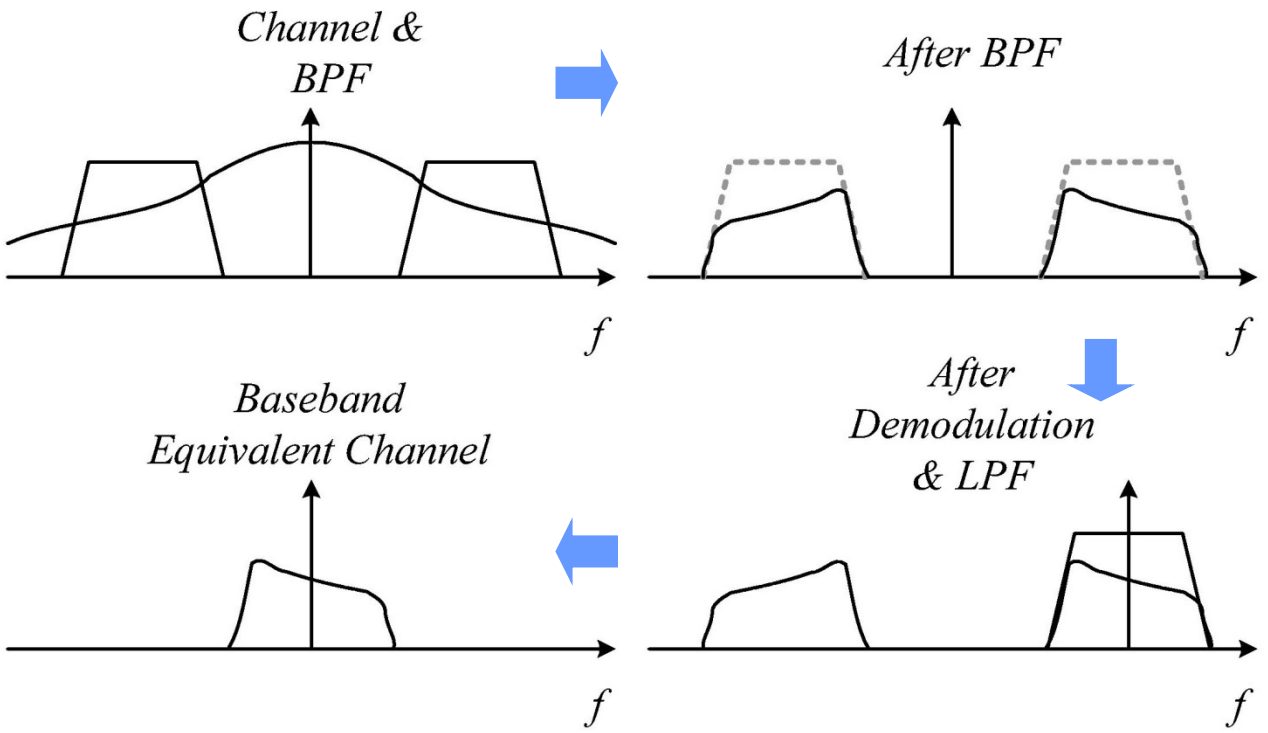
$R(f) = Y(f+f_c)H_{LPF}(f)$

$Y(f) = X(f)H_c(f)H_{BPF}(f)$

Baseband Equivalent Channel (5/5)

- Frequency domain channel response

$$R(f) = \frac{1}{2}\{P(f)H_c(f+f_c)H_{BPF}(f+f_c)H_{LPF}(f)\}$$



Signal Propagation

- Path loss
- Shadowing
- Multipath fading
  - Delay spread
  - Doppler spread
  - Angle spread

Propagation Modeling

- Three basic mechanisms
  - Reflection
  - Diffraction
    - Allow radio signal to propagate around curved surfaces and behind obstacles.
  - Scattering
    - Actual received signal is often stronger than that predicted by reflection and diffraction.
    - When a radio wave is incident on a rough surface, scattering in all directions occurs.

Fading

- Large-scale fading
  - Related with average signal power and SNR.
  - Mainly depending on distance and environments.
- Small-scale fading
  - Challenge to all functional blocks in receivers.
  - Influenced by mobility.
  - Described by channel power delay profile.

Large-Scale Fading – Path Loss

- Energy loss in propagation, related with distance
  - Path loss model

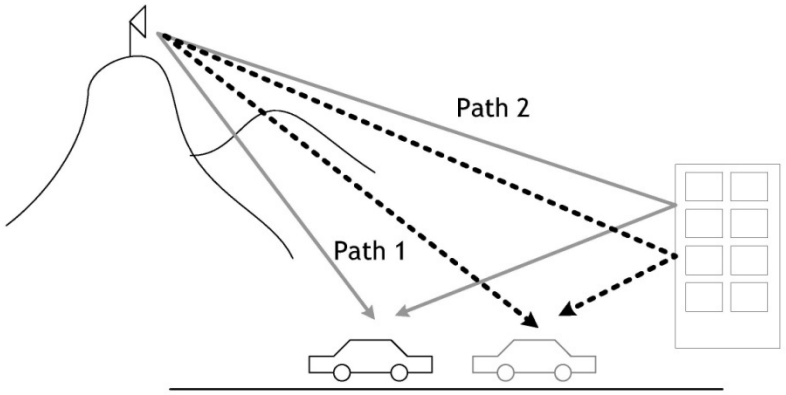
    $PL(d)=PL(d_0)+10n\ log\ (d/d_0)$ (dB).
  - Path loss exponent (*n*) depends on propagation environments
    - Free space, n=2
    - Shadowed urban cellular radio, n=3~5
    - The more obstacles, the larger n is

Large-Scale Fading – Shadowing

- The actual path loss is a random variable (due to random shadowing).
- Log-normal shadowing model
  - $PL(d)=PL(d_0)+10n\ log\ (d/d_0)+X$ (dB)
  - *X* is a zero-mean Gaussian random variable and its variance depends on geographical features of surroundings.

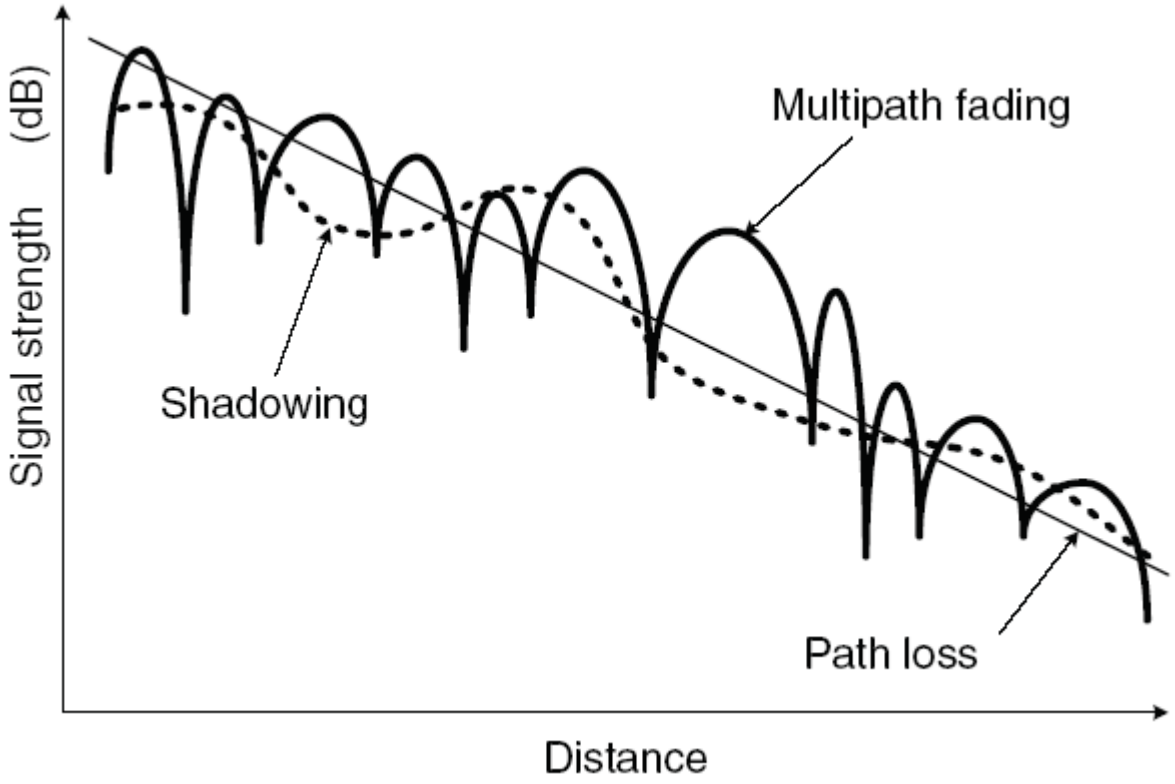
Small-Scale Fading – Multipath Fading

- Multipath fading is the essential concern of wireless transmission.
  - Variation between arrival times of different paths (flat/frequency-selective fading)
  - Relative motion speed between the transmitter and the receiver (slow/fast fading)



Signal Propagation Effect

- Fluctuation of signal strength



Multipath Channel Parameters

- Baseband complex impulse response of a multipath channel

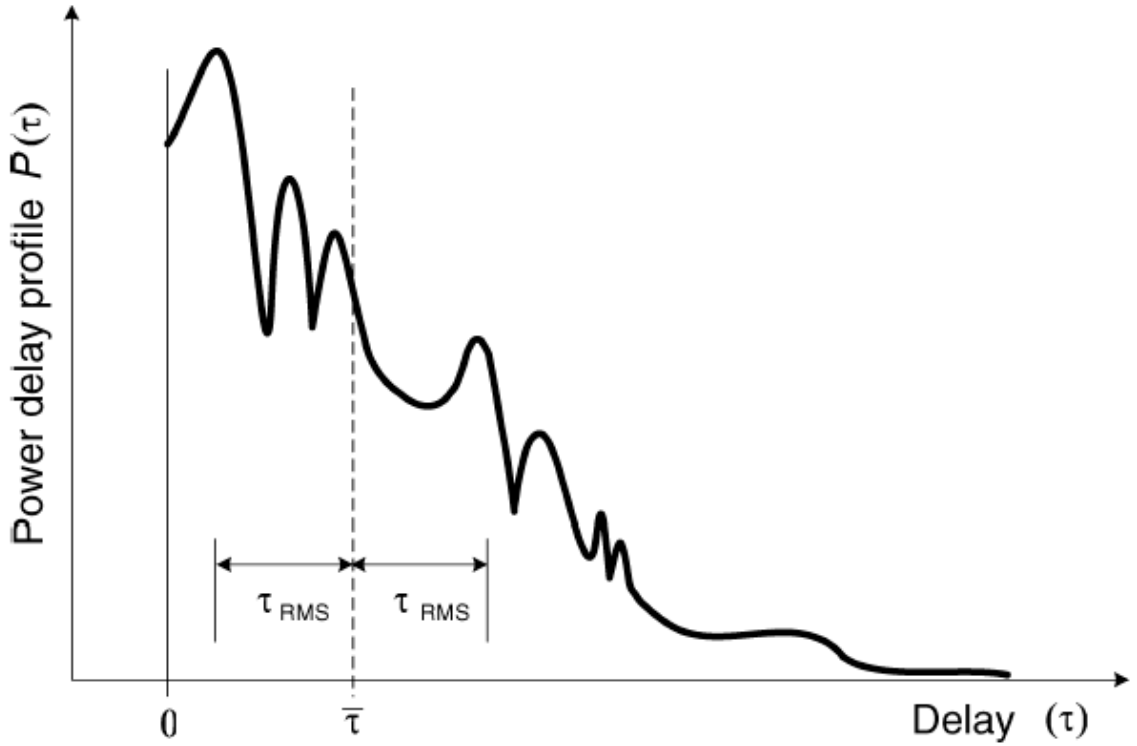
$$h(\tau;t) = \sum_r \beta_r(t) e^{j\theta_r(t)} \delta(\tau - \tau_r)$$

  - Path gain $\beta_r(t)$
  - Phase shift $\theta_r(t)$
  - Time delay τ_r

Channel Power-Delay Profile

- Power delay profile

$$P(\tau;t) \equiv |h(\tau;t)|^2 = \sum_r \beta_r^2(t)\delta(\tau - \tau_r)$$



Delay Spread

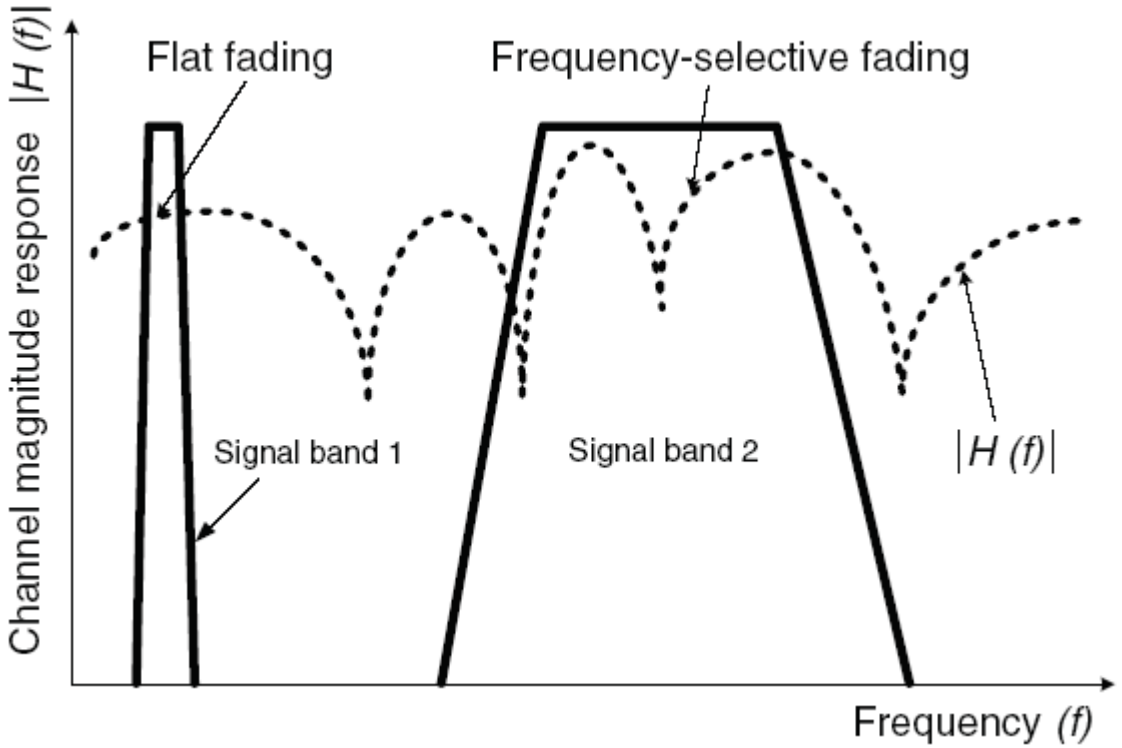
- Mean excess delay $\bar{\tau} = \int_{-\infty}^{\infty} \tau P_{\text{norm}}(\tau) d\tau$
- Normalized PDP $P_{\text{norm}}(\tau) = \frac{P(\tau)}{\int_{-\infty}^{\infty} P(\tau) d\tau}$
- RMS delay spread $\tau_{\text{RMS}} = \sqrt{\int_{-\infty}^{\infty} (\tau - \bar{\tau})^2 P_{\text{norm}}(\tau) d\tau}$

Coherence Bandwidth (1/2)

- The reciprocal of the multipath spread is a measure of the coherence bandwidth.
  - $B_c = 1/(K\tau_{RMS}),\ K=5\sim50$
- Two sinusoids with frequency separation greater than B_c are affected differently by the channel.
- Signal bandwidth v.s. coherence bandwidth
  - Frequency-selective fading
  - Flat fading

Coherence Bandwidth (2/2)

- Flat fading and frequency-selective fading

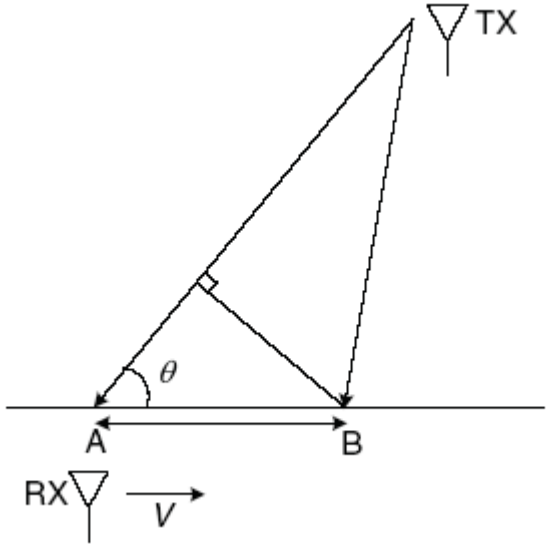


Doppler Effect

- Doppler effect – frequency dispersion

$$f_D = \frac{\Delta\phi}{2\pi\Delta t} = \frac{V\cos\theta}{\lambda}$$

- Maximum Doppler frequency
  - $f_m = \nu/\lambda$,
  - ν is mobility
  - λ is wavelength



$$\Delta\phi = \frac{2\pi\Delta l \cdot \cos\theta}{\lambda} = \frac{2\pi V \cdot \Delta t \cdot \cos\theta}{\lambda}$$

Doppler Spread

- Mean frequency

$$\bar{f} = \int_{-\infty}^{\infty} f S_{\text{norm}}(f) df$$

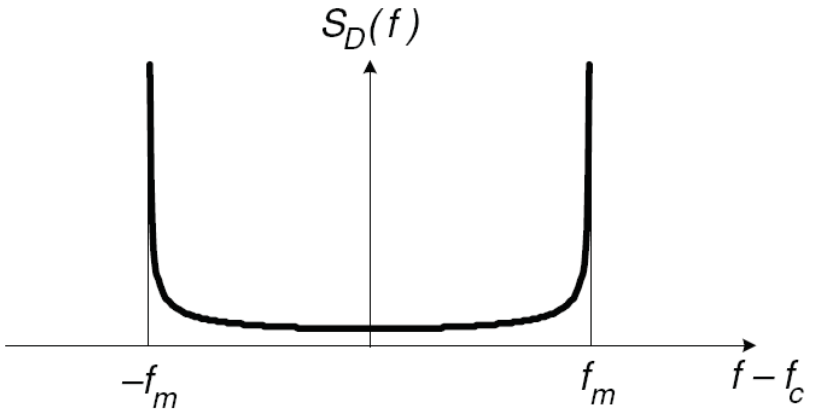
- Normalized spectrum

$$S_{\text{norm}}(f) = \frac{S_D(f)}{\int_{-\infty}^{\infty} S_D(f) df}$$

- RMS Doppler spread

$$B_{\text{RMS}} = \sqrt{\int_{-\infty}^{\infty} (f - \bar{f})^2 S_{\text{norm}}(f) df}$$

Classical Doppler spectrum

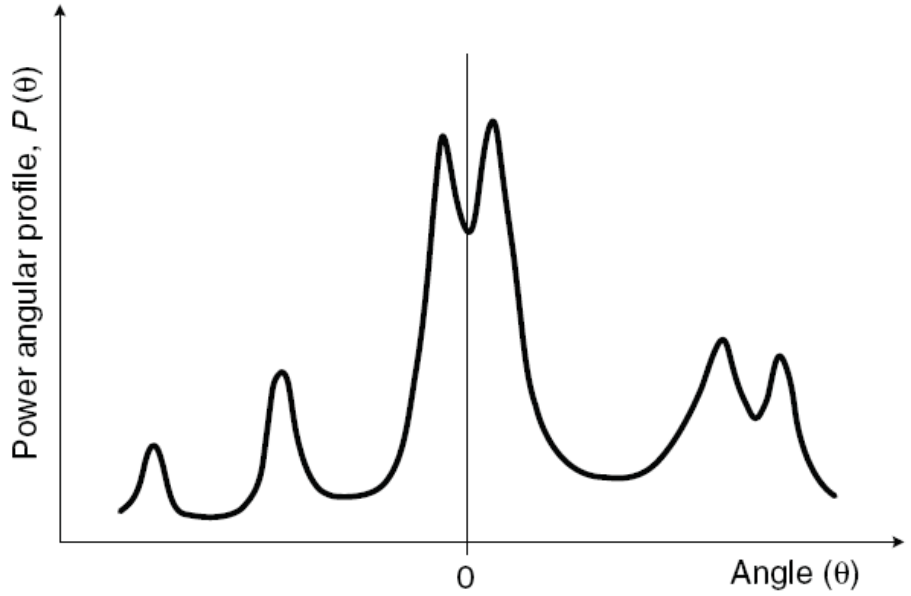


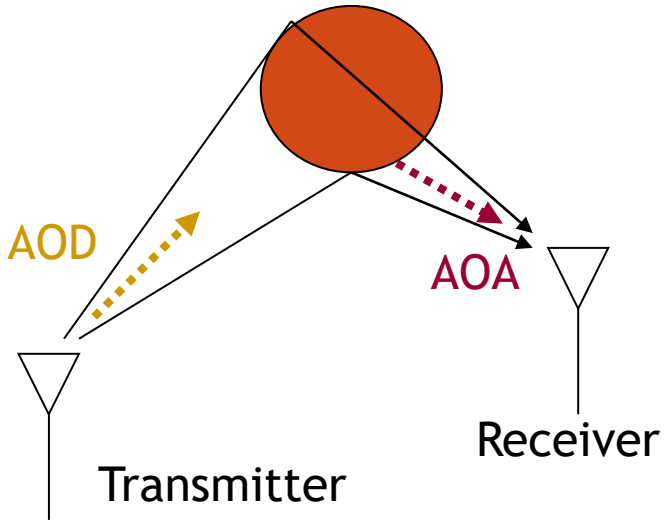
Coherence Time

- The reciprocal of the Doppler spread is a measure of coherence time.
  - $T_c = K/f_m$, f_m is the maximum Doppler frequency
  - Rule of thumb, K=0.423
- A slowly changing channel introduces a large coherence time or a small Doppler spread.
- Symbol time v.s. coherence time
  - Fast fading (Time-selective fading)
  - Slow fading

Power Angular Profile

- Signal spread in spatial angle
  - Angle of departure (AOD)
  - Angle of arrival (AOA)





Angle Spread

- Mean AOA/AOD $$\bar{\theta} = \int_{-\infty}^{\infty} \theta P_{\text{norm}}(\theta) d\theta$$

- Normalized angle power spectrum $$P_{\text{norm}}(\theta) = \frac{P(\theta)}{\int_{-\infty}^{\infty} P(\theta) d\theta}$$

- RMS angle spread $$\theta_{\text{RMS}} = \sqrt{\int_{-\infty}^{\infty} (\theta - \bar{\theta})^2 P_{\text{norm}}(\theta) d\theta}.$$

Coherence Distance

- The degree of space selective fading is characterized by coherence distance, corresponding to angle spread.
- Antenna spacing v.s. coherence distance

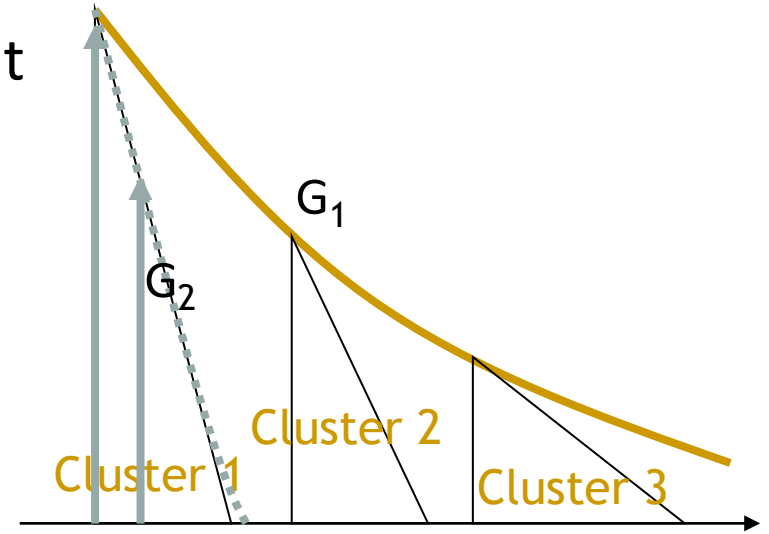
Stationary Rayleigh Fader

- Complex channel impulse response

$$h(\tau) = \sum_{l=0}^{\infty} \sum_{r=0}^{\infty} h_{l,r} \delta(\tau - T_l - \tau_r^{(l)}) = \sum_{l=0}^{\infty} \sum_{r=0}^{\infty} \beta_{r,l} e^{j\theta_{r,l}} \delta(\tau - T_l - \tau_r^{(l)}).$$

- Double exponential decay
  - Cluster power decay time constant
    - Γ_1
  - Path power decay time constant
    - Γ_2

$$\overline{\beta_{r,l}^2} = \beta^2 e^{-T_l/\Gamma_1} e^{-\tau_{r,l}/\Gamma_2}$$



Double exponential decay

Example: Channel Model for Rural Area

- 3GPP Channel model

Tap	Delay (μs)	Average relative power (dB)	Doppler spectrum
1	0	−5.2	Direct path $f_D = 0.7f_m$
2	0.042	−6.4	Classical
3	0.101	−8.4	Classical
4	0.129	−9.3	Classical
5	0.149	−10.0	Classical
6	0.245	−13.1	Classical
7	0.312	−15.3	Classical
8	0.410	−18.5	Classical
9	0.469	−20.4	Classical
10	0.528	−22.4	Classical

Front-end Electronics Effect

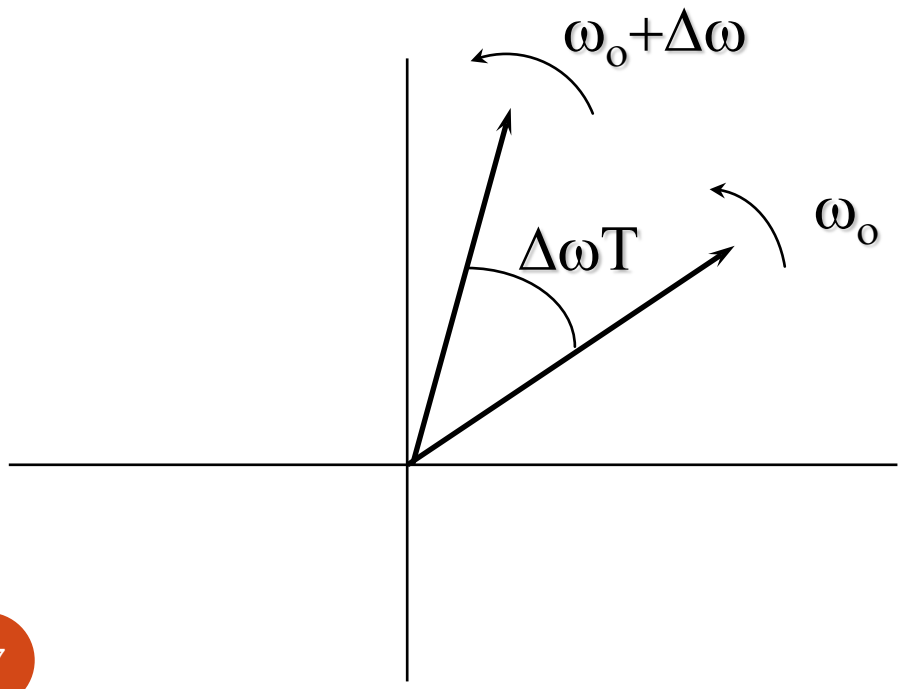
Front-end Electronics Effect

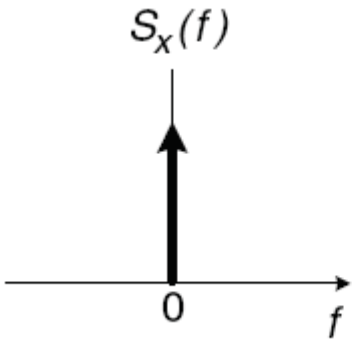
- Carrier frequency offset
- Sampling clock offset
- Phase noise
- IQ imbalance
- DC offset
- PA non-linearity

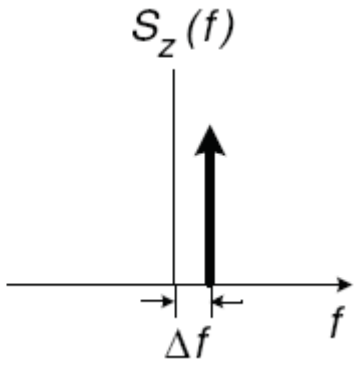
Carrier Frequency Offset

- Oscillator mismatch and Doppler frequency cause the carrier frequency offset.

$$x(t) = \mathrm{Re}\{x_{BB}(t)e^{j2\pi f_c t}\} = \mathrm{Re}\{x_{BB}(t)e^{-j2\pi\Delta f t}e^{j2\pi(f_c+\Delta f)t}\}$$





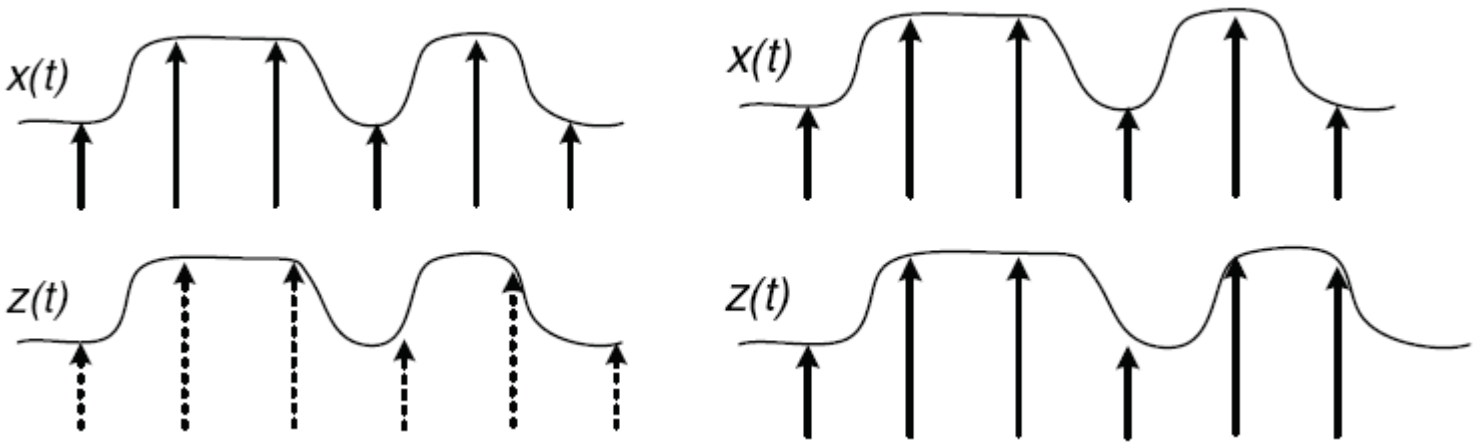


Sampling Clock Offset

- Oscillator mismatch and Doppler frequency cause the sampling clock offset.

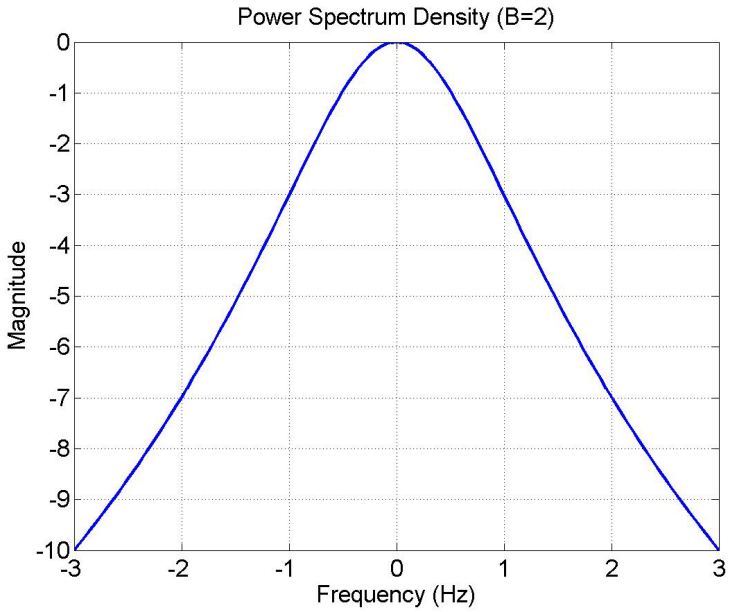
$$x(t)\Big|_{t=nT_s} = \mathrm{Re}\{x_{BB}(t)e^{j2\pi f_c t}\}\Big|_{t=nT_s}$$

$$x\left(t-\frac{\Delta l}{c}\right)\Big|_{t=(n+1)T_s} = \mathrm{Re}\left\{x_{BB}\left(t-\frac{\Delta l}{c}\right)e^{j2\pi f_c\left(t-\frac{\Delta l}{c}\right)}\right\}\Big|_{t=(n+1)T_s}$$



Phase Noise

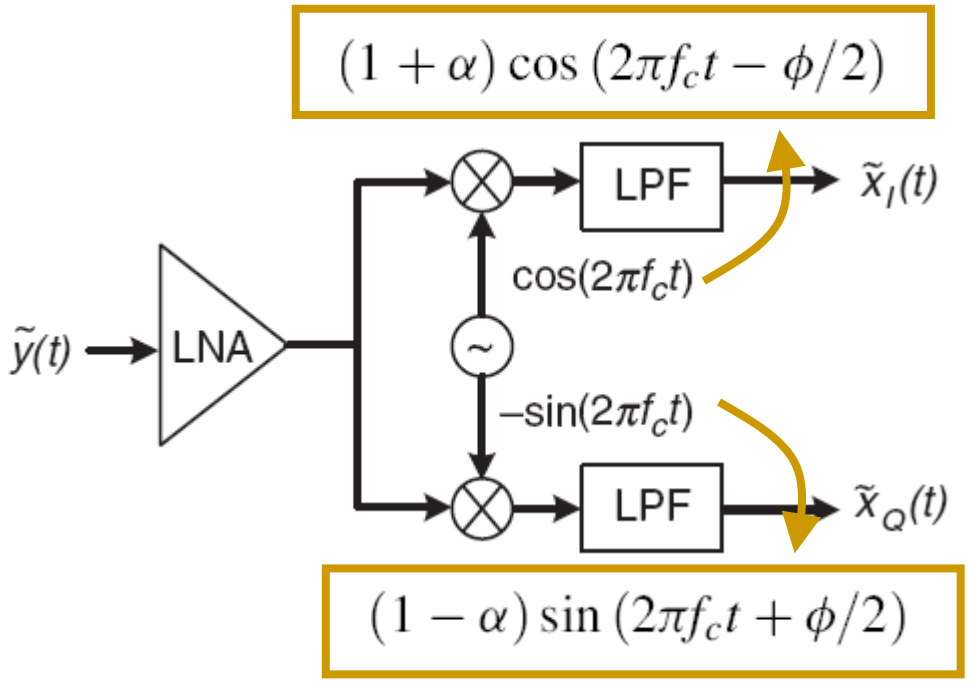
- Oscillator generates phase noise, which produces a random phase rotation.
  - Modeled as a discrete-time Wiener-Levy process
  - *B*: two-side 3-dB bandwidth.



$$S(f) = \frac{2}{\pi B} \frac{1}{\left[1 + \left(\frac{2f}{B}\right)^2\right]},$$

I/Q Imbalance

- Gain imbalance $20\log\frac{1+\alpha}{1-\alpha}$ dB
- Phase imbalance ϕ

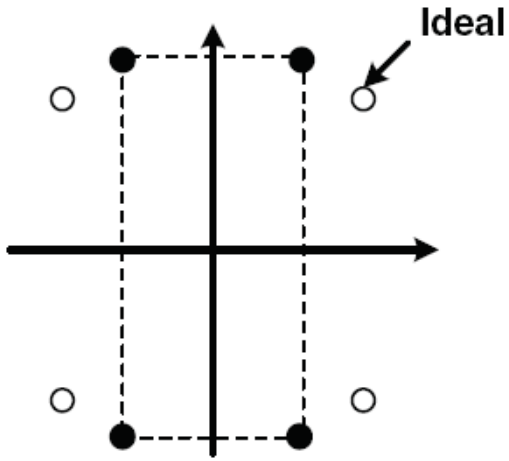


IQ Imbalance Effect

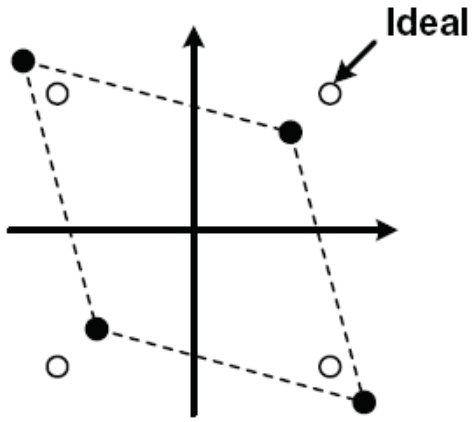
- QPSK signals

$$\tilde{x}_I(t) = (1+\alpha)\left[x_I(t)\cos\left(\frac{\phi}{2}\right) - x_Q(t)\sin\left(\frac{\phi}{2}\right)\right]$$

$$\tilde{x}_Q(t) = (1-\alpha)\left[x_Q(t)\cos\left(\frac{\phi}{2}\right) - x_I(t)\sin\left(\frac{\phi}{2}\right)\right]$$



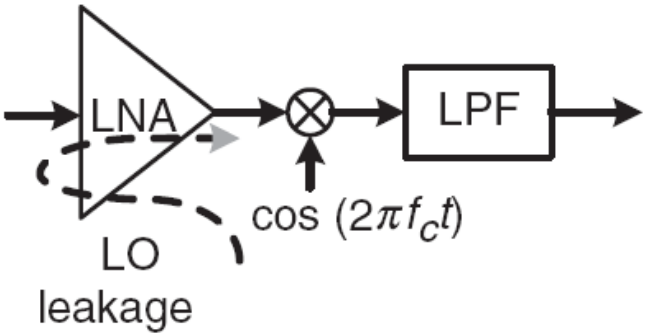
Gain Error

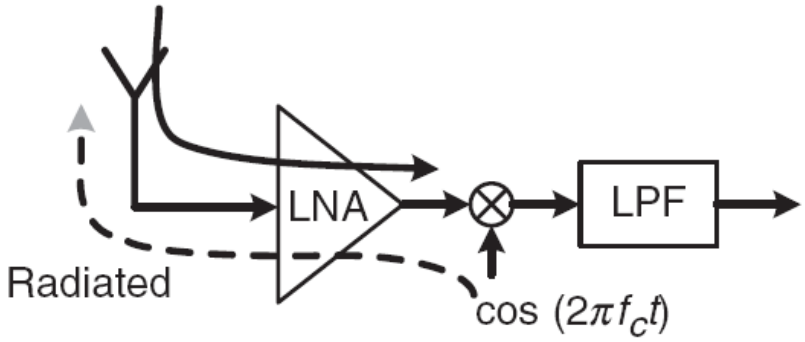


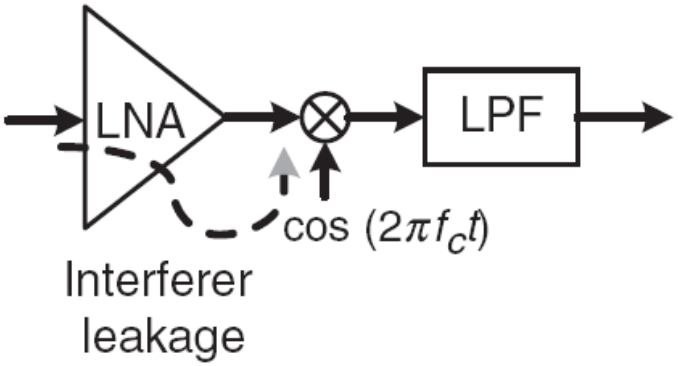
Phase Error

DC Offset

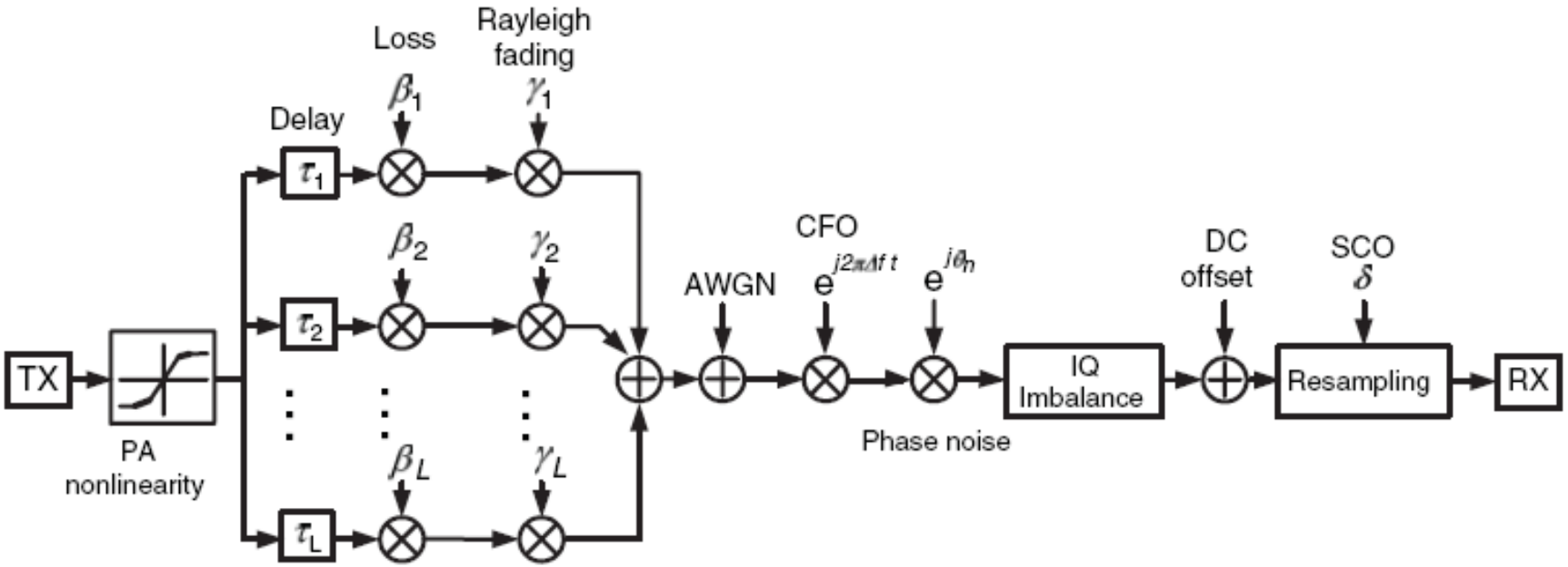
- Self-mixing problem
  - LO leakage
  - LO re-radiation
  - Interference leakage







Equivalent Baseband Channel Model



Rayleigh Fading

- If the envelope of channel impulse response g_r is modeled as a zero-mean complex-value Gaussian process, the channel is said to be a Rayleigh fading channel.
  - Mean, $E\{g_r\}=0$
  - $|\ g_r\ |$, Rayleigh distribution
  - $\angle g_r$, uniform distribution

Rician Fading

- If there is one strong signal coming from a direct path, the channel impulse response can be modeled as a non zero-mean Gaussian process, it is Rician fading.
  - $|g_r|$, Rician distribution

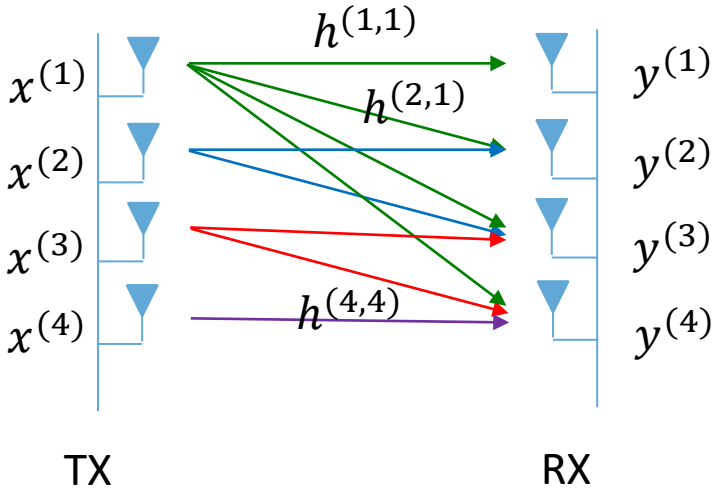
Three-Dimensional mmWave MIMO Channel Model

Outline

- MIMO channel model
- Uniform linear array and uniform planar array for beamforming
- Three-dimensional channel model
- mmWave parameters

MIMO Channel Response (1/2)

- MIMO **flat fading** channel
  - If channel response is **independent**, the elements in $\mathbf{H}$ are described by independent and identically-distributed (i.i.d.) Gaussian random variables ($\mathbf{H}_{iid}$) with zero mean and unit variance.



$$\mathbf{H} = \begin{bmatrix} h^{(1,1)} & h^{(1,2)} & h^{(1,3)} & h^{(1,4)} \\ h^{(2,1)} & h^{(2,2)} & h^{(2,3)} & h^{(2,4)} \\ h^{(3,1)} & h^{(3,2)} & h^{(3,3)} & h^{(3,4)} \\ h^{(4,1)} & h^{(4,2)} & h^{(4,3)} & h^{(4,4)} \end{bmatrix}$$

$$\mathbf{y} = \begin{bmatrix} y^{(1)} \\ y^{(2)} \\ y^{(3)} \\ y^{(4)} \end{bmatrix} = \mathbf{Hx} + \mathbf{n} = \begin{bmatrix} h^{(1,1)} & h^{(1,2)} & h^{(1,3)} & h^{(1,4)} \\ h^{(2,1)} & h^{(2,2)} & h^{(2,3)} & h^{(2,4)} \\ h^{(3,1)} & h^{(3,2)} & h^{(3,3)} & h^{(3,4)} \\ h^{(4,1)} & h^{(4,2)} & h^{(4,3)} & h^{(4,4)} \end{bmatrix} \begin{bmatrix} x^{(1)} \\ x^{(2)} \\ x^{(3)} \\ x^{(4)} \end{bmatrix} + \begin{bmatrix} n^{(1)} \\ n^{(2)} \\ n^{(3)} \\ n^{(4)} \end{bmatrix}$$

MIMO Channel Response (2/2)

- MIMO **flat fading** channel
  - If channel response is **correlated**, the elements in $\mathbf{H}$ can be generated by the following method

$$\mathbf{H} = [\mathbf{R}_{rx}]^{1/2}\mathbf{H}_{iid}\left([\mathbf{R}_{tx}]^{1/2}\right)^{T}$$

  - $[\mathbf{R}_{tx}]$ and $[\mathbf{R}_{rx}]$ are transmit and receive correlation matrices
  - The (i,l)th elements in $\mathbf{R}_{tx}$ and $\mathbf{R}_{rx}$ are $\rho_{tx,i,l}$ and $\rho_{rx,i,l}$, which are the complex correlation coefficients between i th and l th antennas

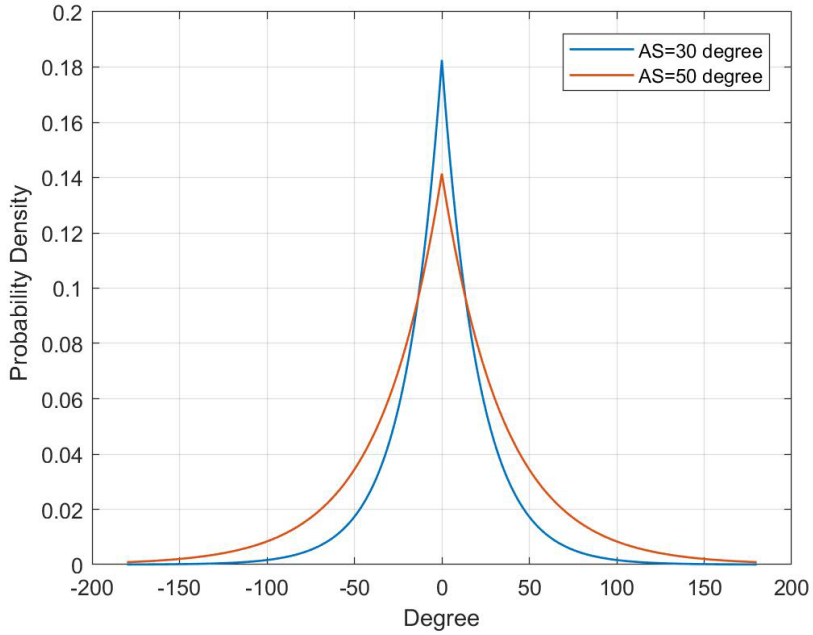
$$\mathbf{R}_{tx} = \begin{bmatrix} 1 & \rho_{tx,1,2} & \rho_{tx,1,3} & \rho_{tx,1,4} \\ \rho_{tx,2,1} & 1 & \rho_{tx,2,3} & \rho_{tx,2,4} \\ \rho_{tx,3,1} & \rho_{tx,3,2} & 1 & \rho_{tx,3,4} \\ \rho_{tx,4,1} & \rho_{tx,4,2} & \rho_{tx,4,3} & 1 \end{bmatrix}$$

Power Angular Spectrum (PAS)

- The angle of arrival statistics within a cluster can be described by Laplacian distribution [3]

$$p(\theta) = \frac{1}{\sqrt{2}\sigma} e^{-|\sqrt{2}\theta/\sigma|}$$

- σ is the standard deviation of the PAS corresponding to the angle spread (AS).



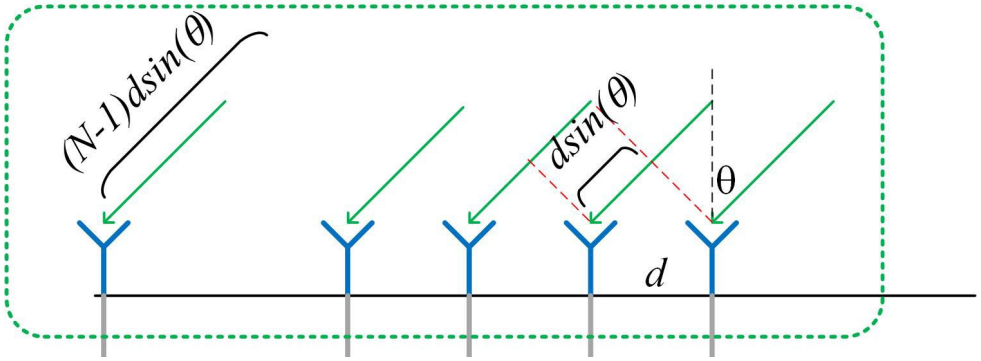
Channel Correlation (1/2)

- Given the PAS, the channel correlation between antenna i and antenna l can be described by

$$\rho_{tx,i,l} = E_\theta\{e^{j\phi_{i,l}}\} = E_\theta\left\{\cos(\frac{2\pi}{\lambda} d_{i,l}\sin(\theta)) + j\sin(\frac{2\pi}{\lambda} d_{i,l}\sin(\theta))\right\}$$

$$= R_{XX} + jR_{XY}$$

$$= \int_{-\pi}^{\pi}\left(\cos\left(\frac{2\pi}{\lambda} d_{i,l}\sin(\theta)\right) + j\sin\left(\frac{2\pi}{\lambda} d_{i,l}\sin(\theta)\right)\right)p(\theta)d\theta$$



Channel Correlation (2/2)

- If no PAS, but channel correlation is desired, several kinds of correlation matrices are used
  - Use correlation coefficient directly [3]

$$\mathbf{R}_{tx} = \begin{bmatrix} 1 & c_{tx} & \cdots & c_{tx}^{(N_t-1)^2} \\ c_{tx} & 1 & \cdots & c_{tx}^{(N_t-2)^2} \\ \vdots & \vdots & \ddots & \vdots \\ c_{tx}^{(N_t-1)^2} & c_{tx}^{(N_t-2)^2} & \cdots & 1 \end{bmatrix}$$

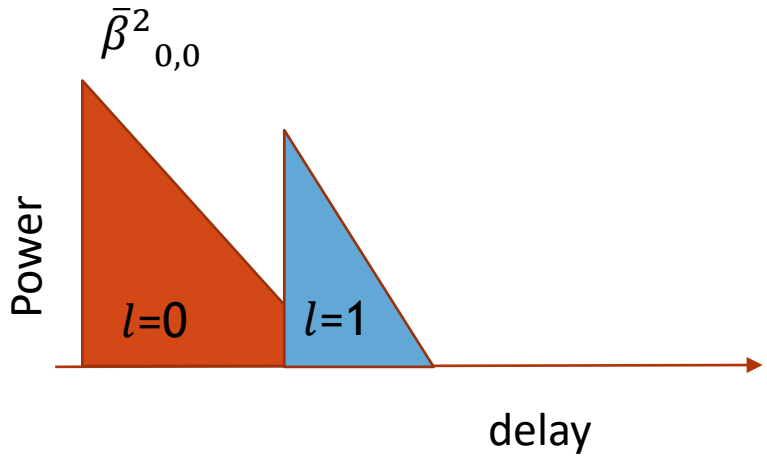
  - Use distance $d_{i,l}$ [4], $\rho_{tx,i,l} = sinc(\frac{2\pi}{\lambda} d_{i,l})$

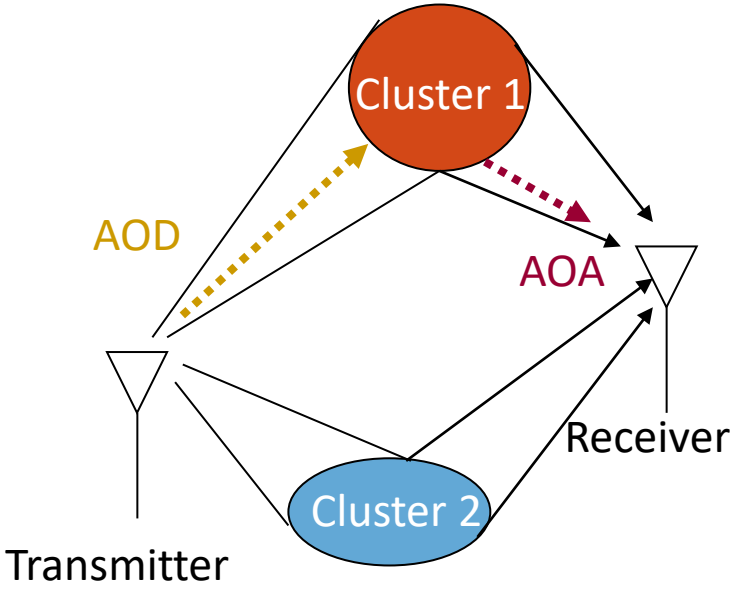
$$\mathbf{R}_{tx} = \begin{bmatrix} 1 & \rho_{tx,1,2} & \cdots & \rho_{tx,1,N_t} \\ \rho_{tx,2,1} & 1 & \cdots & \rho_{tx,2,N_t} \\ \vdots & \vdots & \ddots & \vdots \\ \rho_{tx,N_t,1} & \rho_{tx,N_t,2} & \cdots & 1 \end{bmatrix}$$

Saleh-Valenzuela Model

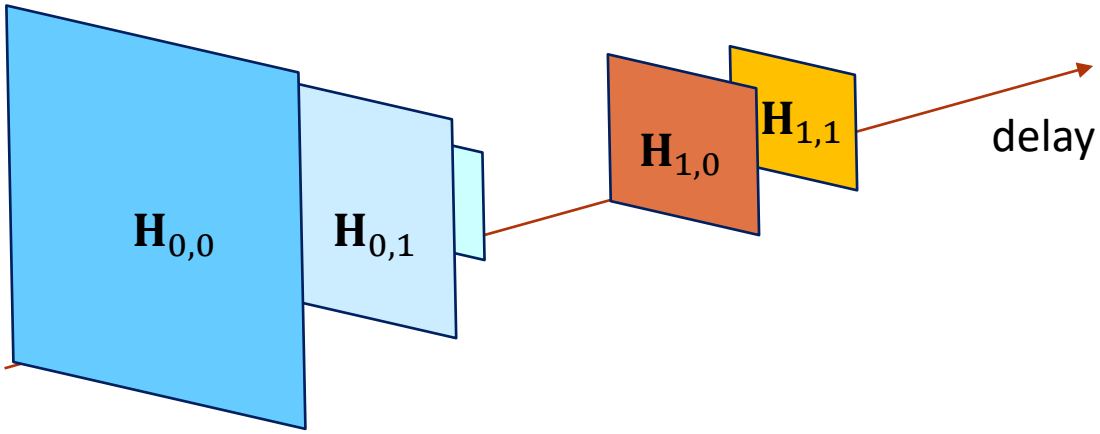
- Cluster model approach by Saleh and Valenzuela

$$p_{l,m} = \bar{\beta}^2{}_{l,m} = \bar{\beta}^2{}_{0,0} e^{-T_l/\Gamma} e^{-\tau_{l,m}/\gamma}$$





MIMO Frequency-Selective Fading Channel Model



$$\mathbf{H}(\tau) = \sum_{l=0}^{L-1} \sum_{m=0}^{M-1} \sqrt{p_{l,m}} \mathbf{H}_{l,m} \delta(\tau - T_l - \tau_{l,m})$$

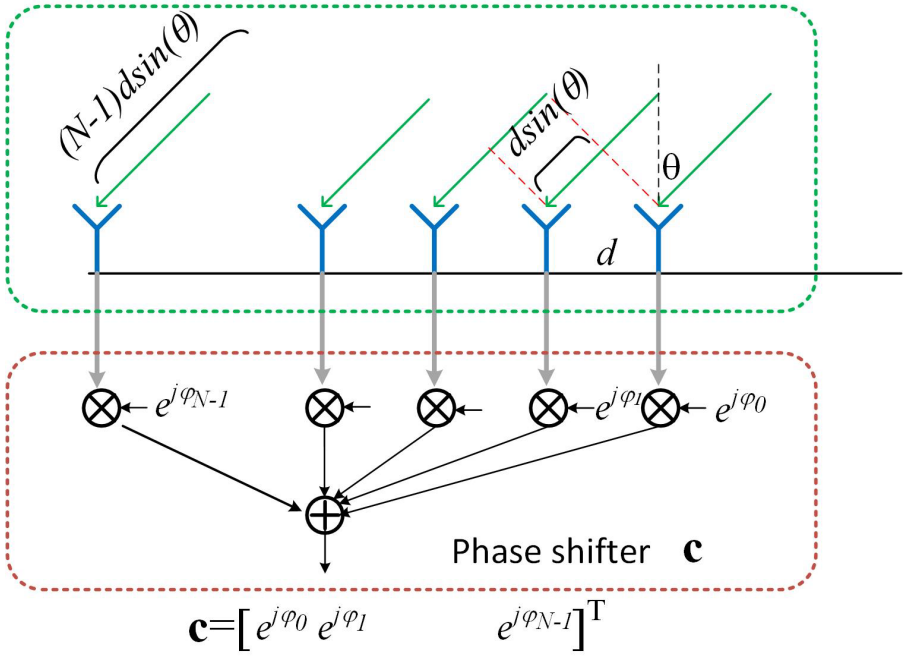
$$\mathbf{H}_{l,m} = \begin{bmatrix} h_{l,m}^{(1,1)} & h_{l,m}^{(1,2)} & h_{l,m}^{(1,3)} & h_{l,m}^{(1,4)} \\ h_{l,m}^{(2,1)} & h_{l,m}^{(2,2)} & h_{l,m}^{(2,3)} & h_{l,m}^{(2,4)} \\ h_{l,m}^{(3,1)} & h_{l,m}^{(3,2)} & h_{l,m}^{(3,3)} & h_{l,m}^{(3,4)} \\ h_{l,m}^{(4,1)} & h_{l,m}^{(4,2)} & h_{l,m}^{(4,3)} & h_{l,m}^{(4,4)} \end{bmatrix}$$

IEEE TGn Channel Model B [5]

	Tap index	1	2	3	4	5	6	7	8	9
	Excess delay [ns]	0	10	20	30	40	50	60	70	80
Cluster 1	Power [dB]	0	-5.4	-10.8	-16.2	-21.7				
AoA	AoA [°]	4.3	4.3	4.3	4.3	4.3				
AS (receiver)	AS [°]	14.4	14.4	14.4	14.4	14.4				
AoD	AoD [°]	225.1	225.1	225.1	225.1	225.1				
AS (transmitter)	AS [°]	14.4	14.4	14.4	14.4	14.4				
Cluster 2	Power [dB]			-3.2	-6.3	-9.4	-12.5	-15.6	-18.7	-21.8
AoA	AoA [°]			118.4	118.4	118.4	118.4	118.4	118.4	118.4
AS	AS [°]			25.2	25.2	25.2	25.2	25.2	25.2	25.2
AoD	AoD [°]			106.5	106.5	106.5	106.5	106.5	106.5	106.5
AS	AS [°]			25.4	25.4	25.4	25.4	25.4	25.4	25.4

Antenna Array for Beamforming

Uniform Linear Antenna Array

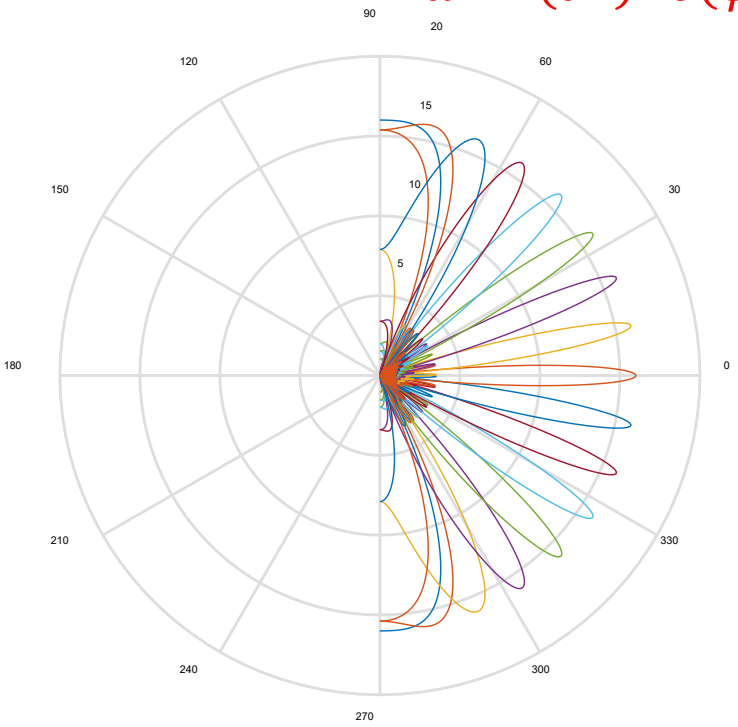


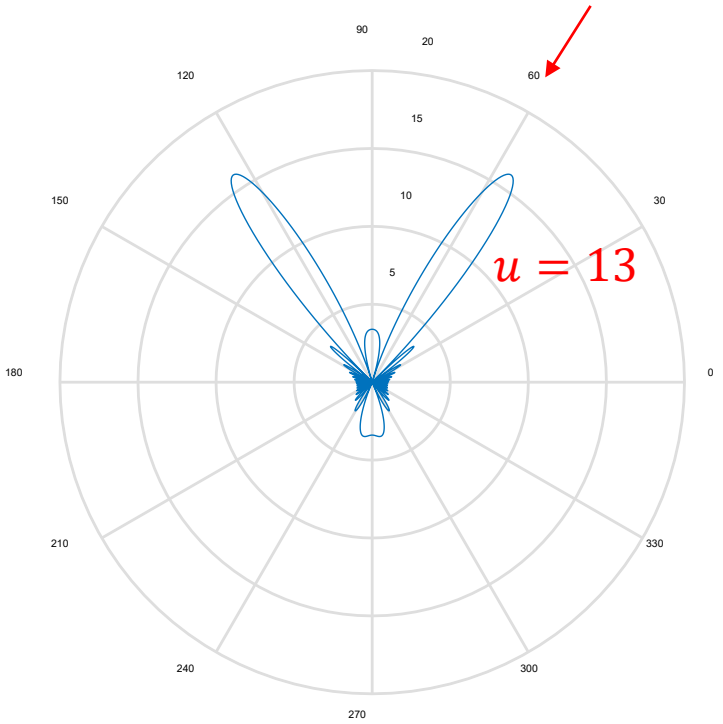
$$\boldsymbol{a}^{\boldsymbol{ULA}}(\theta) = [1 \quad e^{-jkdsin(\theta)} \quad \dots \quad e^{-j(N-1)kdsin(\theta)}]^T \quad k = \frac{2\pi}{\lambda}$$

$$\mathbf{H} = \frac{1}{\sqrt{L}} \sum_{l=1}^{L} \alpha_l \Lambda^r(\theta_l^r)\, \Lambda^t(\theta_l^t) \boldsymbol{a}^{\boldsymbol{ULA,r}}(\theta_l^r) \boldsymbol{a}^{\boldsymbol{ULA,t}}(\theta_l^t)^H$$

2D Beamforming (1/2)

- Assume $\Lambda(\theta) = 1$, single path, $N = 16$ and $\theta^r = \pi/3$
- After combining $\boldsymbol{a^{ULA}(\theta^r)^H c(\varphi)}$





$\boldsymbol{c}(\varphi) = [1 \quad e^{-jkdsin(\varphi)} \quad \dots \quad e^{-j(N-1)kdsin(\varphi)}]^{\mathrm{T}}$ $\varphi_u = \dfrac{(u-8)\pi}{16}$

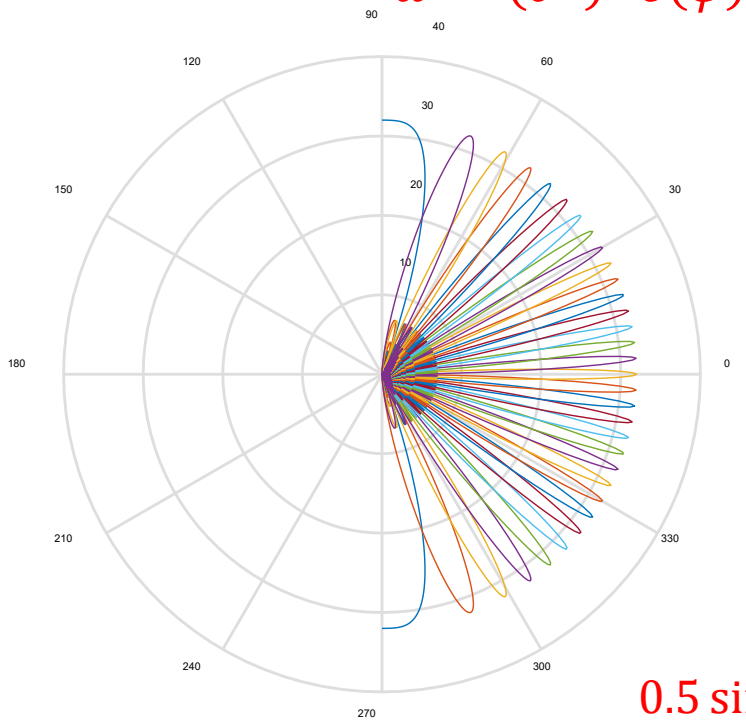
$u = 0,1, \dots N-1$

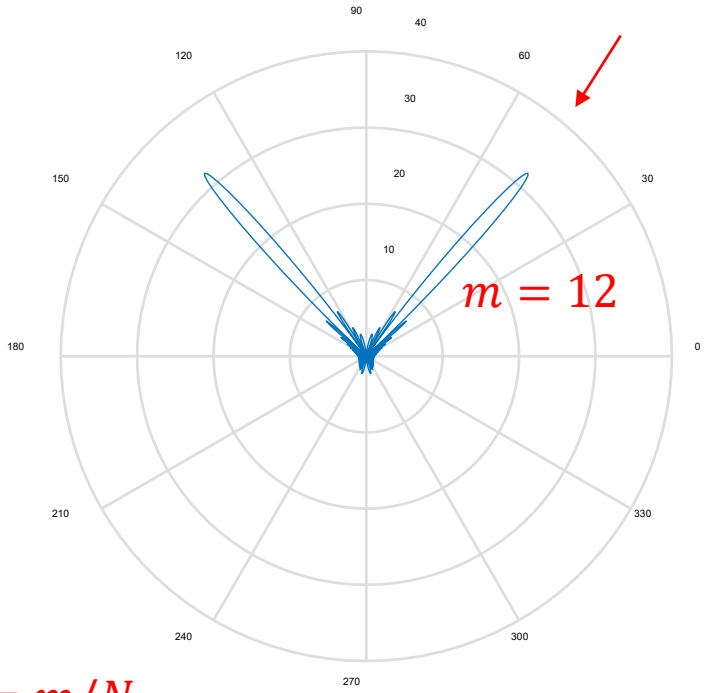
φ_u: the uth beam in the codebook

2D Beamforming (2/2)

- Assume $\Lambda(\theta) = 1$, single path, $N = 32$ and $\theta^r = \sin^{-1}(0.75) = 48.59^\circ$ After combining

$$\boldsymbol{a}^{\boldsymbol{ULA}}(\theta^r)^{\boldsymbol{H}}\boldsymbol{c}(\varphi)$$





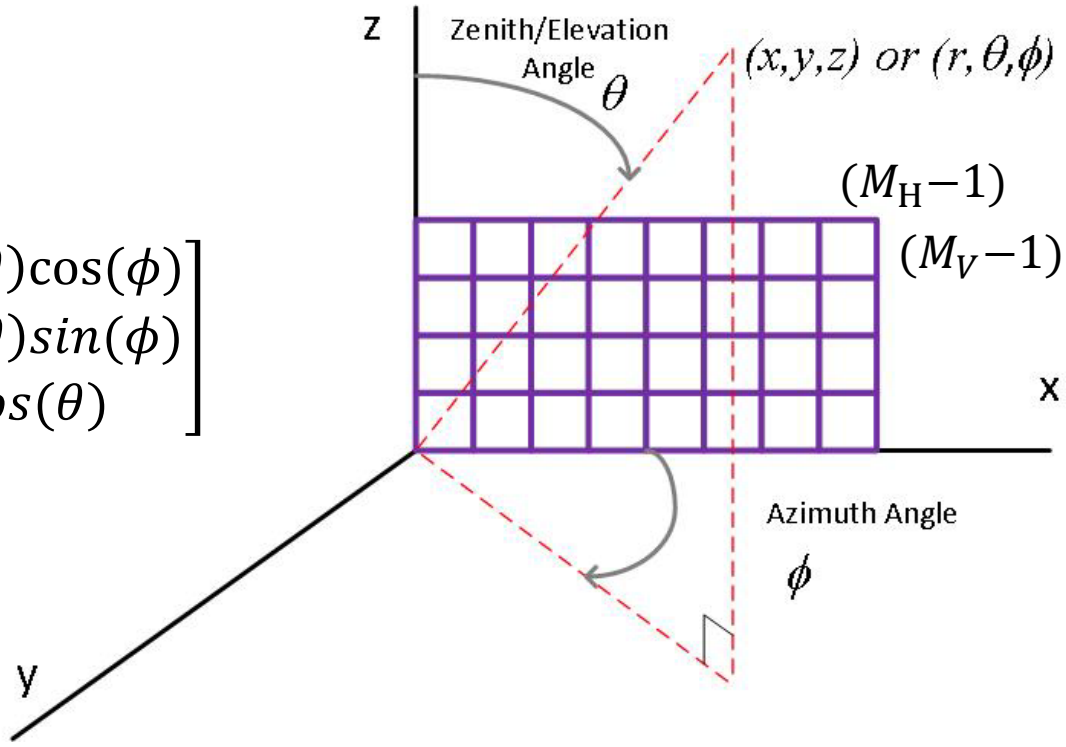
$$0.5\sin(\varphi) = m/N$$

$$\boldsymbol{c}(\varphi) = [1 \quad e^{j2\pi m/N} \quad \dots \quad e^{j2\pi(N-1)m/N}]^{\mathbf{T}}$$

$$\varphi'_m = \sin^{-1}\left(\frac{2m}{N}\right)$$

$$m = -16, -15, \dots, 15$$

Planar Antenna Array



$$\begin{bmatrix} sin(\theta)\cos(\phi) \\ sin(\theta)sin(\phi) \\ cos(\theta) \end{bmatrix}$$

$$\begin{bmatrix} m_H d \\ 0 \\ m_V d \end{bmatrix}$$

$$\boldsymbol{a}^{\boldsymbol{UPA}}(\theta) = \begin{bmatrix} 1 & e^{jkd(m_H sin(\theta)\cos(\phi)+m_V\cos(\theta))} & \dots & e^{jkd\left((M_H-1)sin(\theta)\cos(\phi)+(M_V-1)\cos(\theta)\right)} \end{bmatrix}^T$$

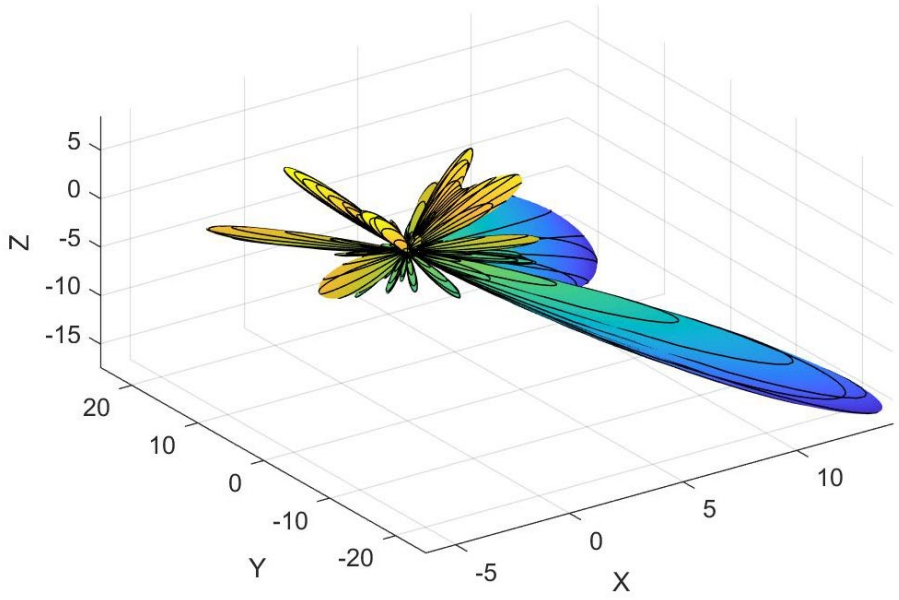
$$\mathbf{H} = \frac{1}{\sqrt{L}}\sum_{l=1}^{L} \alpha_l \Lambda^r(\theta_l^r,\phi_l^r)\,\Lambda^t(\theta_l^t,\phi_l^t)\boldsymbol{a}^{\boldsymbol{UPA,r}}(\theta_l^r,\phi_l^r)\boldsymbol{a}^{\boldsymbol{UPA,t}}(\theta_l^t,\phi_l^t)^H$$

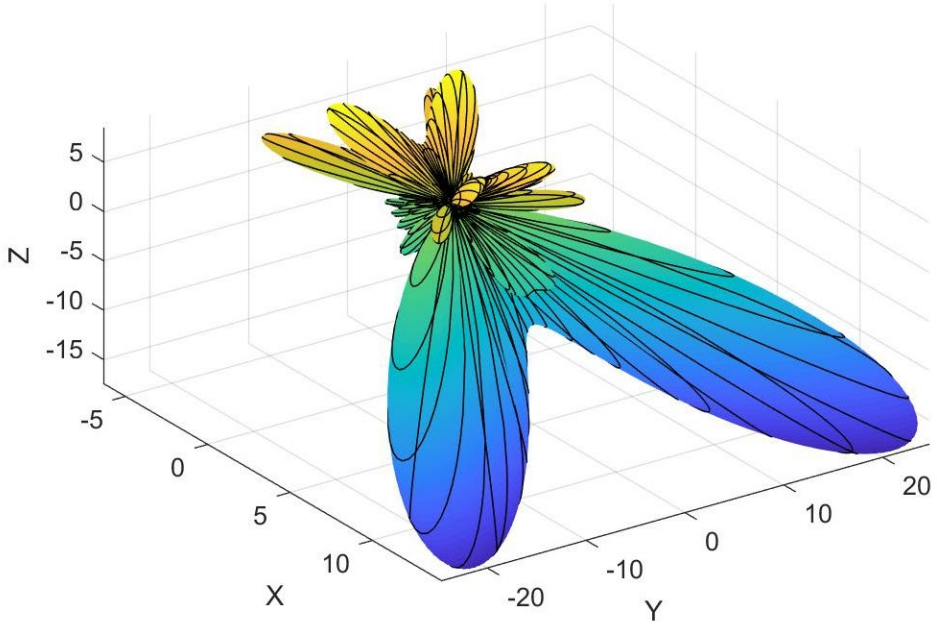
3D Beamforming

- Assume $\Lambda(\theta, \phi) = 1$, single path, and

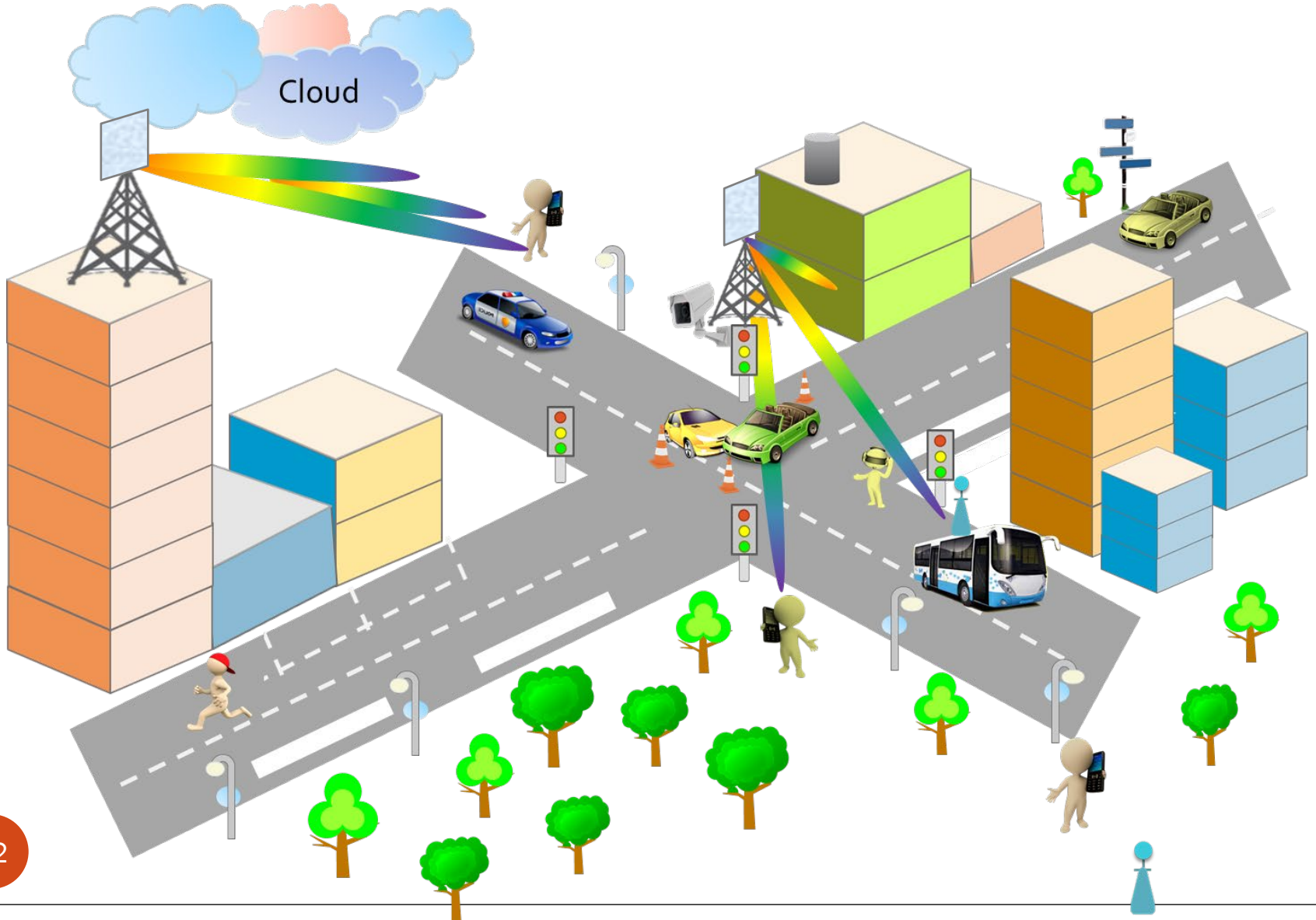
$$\theta^r = \frac{2\pi}{3}, \phi^r = \frac{\pi}{3}$$

- After combining, $\boldsymbol{a}^{\boldsymbol{UPA}}(\theta^r, \phi^r)\boldsymbol{c}(\theta, \phi)^{\boldsymbol{H}}$





3D Beamforming Scenario



Advantages of Beamforming

- Advantages of (digital/analog) beamforming
  - Strengthen signal power
  - Suppress interference

- Note that power is inversely proportional to distance square.
- If power is constrained, how to use power efficiently?
  - Power Efficiency

- If higher resolution in the spatial domain can be achieved, signal-to-noise-and-interference ratio can be enhanced. Capacity is also increased.
  - Spectral Efficiency